

Runge Kutta

$$\begin{array}{c|c} \mathbf{c} & \mathfrak{A} \\ \hline & \mathbf{b}^\top \end{array} = \begin{array}{c|ccc} c_1 & a_{11} & \cdots & a_{1s} \\ \vdots & \vdots & \ddots & \vdots \\ c_s & a_{s1} & \cdots & a_{ss} \\ \hline & b_1 & \cdots & b_s \end{array} \quad \mathbf{c}, \mathbf{b} \in \mathbb{R}^s, \mathfrak{A} \in \mathbb{R}^{s \times s}$$

$$\mathbf{k}_i = \mathbf{f}(t + c_i\tau, \mathbf{u} + \tau \sum_{j=1}^s a_{ij}\mathbf{k}_j) \quad \Psi^{t, t+\tau} \mathbf{u} = \mathbf{u} + \tau \sum_{i=1}^s b_i \mathbf{k}_i$$

$$c_i = \sum_j a_{ij}$$

Explicit Runge Kutta

$$a = \begin{bmatrix} 0 & \cdots & 0 & 0 \\ a_{21} & \cdots & 0 & 0 \\ \vdots & \ddots & \vdots & \vdots \\ a_{s1} & \cdots & a_{ss-1} & 0 \end{bmatrix}$$

$$M(t) \frac{\partial u}{\partial t} = f(t, u)$$

- $M \in \mathbb{R}^{n \times n}$
- $f(t, u) \in \mathbb{R}^n$
- $u \in \mathbb{R}^n$

$$M_{i+1} \mathbf{k}_{i+1} = f \left(t + c_i\tau, u + \tau \sum_{j=1}^j \mathbf{k}_j \right)$$

$$du = \tau \sum_i b_i \mathbf{k}_i$$

Implicit Linear Runge Kutta

$$M(t) \frac{\partial u}{\partial t} = A(t)u + B(t)$$

- $M \in \mathbb{R}^{n \times n}$
- $A \in \mathbb{R}^{n \times n}$
- $B \in \mathbb{R}^n$
- $u \in \mathbb{R}^n$

$$\begin{bmatrix} M_0 - A_0 c_0 \tau a_{0,0} & -A_0 c_1 \tau a_{0,1} & \cdots & -A_0 c_{n-1} \tau a_{0,n-1} \\ -A_1 c_0 \tau a_{1,0} & M_1 - A_1 c_1 \tau a_{1,1} & \cdots & -A_1 c_{n-1} \tau a_{1,n-1} \\ \vdots & \vdots & \ddots & \vdots \\ -A_{n-1} c_0 \tau a_{n-1,0} & -A_{n-1} c_1 \tau a_{n-1,1} & \cdots & M_{n-1} - A_{n-1} c_{n-1} \tau a_{n-1,n-1} \end{bmatrix} \begin{bmatrix} \mathbf{k}_0 \\ \mathbf{k}_1 \\ \vdots \\ \mathbf{k}_{n-1} \end{bmatrix} = \begin{bmatrix} B_0 + A_0 u \\ B_1 + A_1 u \\ \vdots \\ B_{n-1} + A_{n-1} u \end{bmatrix}$$

$$du = \tau \sum_i b_i \mathbf{k}_i$$